AWS VPC PEERING PROJECT

Project Report

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Program	AWS Internship / EWB Student Project
AWS Region	US East (N. Virginia) — us-east-1
Project	VPC Peering between two Amazon VPCs

This report documents the VPC configuration, peering routes, and connectivity validation performed in AWS.

1. Project Overview

The objective of this lab is to connect two separate Amazon Virtual Private Clouds (VPCs) using VPC Peering and configure routes so private instances can communicate across the VPC boundary. A bastion host in VPC A is used for administrative SSH access to the private instances.

2. Network Architecture

Resource	VPC A	VPC B
VPC name	ewb-vpc-a	ewb-vpc-b
IPv4 CIDR	10.10.0.0/16	10.20.0.0/16
Private subnet	10.10.2.0/24	10.20.1.0/24
Private instance	App-A — 10.10.2.10	App-B — 10.20.1.10
Peering connection	ewb-a-b-peer	ewb-a-b-peer

The VPC peering connection ID shown in the AWS console is **pcx-0288c0d3ed98acc43** and its status is Active.

3. Routing Configuration

The VPC A private route table includes a route for destination 10.20.0.0/16 through the peering connection. The VPC B private route table includes a route for destination 10.10.0.0/16 through the same peering connection. Both routes were shown as Active in the supplied AWS console screenshots.

4. Security and Access

A bastion host in the public subnet of VPC A provides SSH access to the private instances. Security group rules were configured for administrative SSH access and ICMP testing between the application instances. The bastion host has private IP 10.10.1.10; App-A has private IP 10.10.2.10; App-B has private IP 10.20.1.10.

5. Validation Results

The captured terminal evidence includes successful ICMP connectivity between App-A and App-B, with the App-B-to-App-A test showing 4 packets transmitted, 4 received, and 0% packet loss. The screenshots also document SSH access through the bastion host and the active peering connection and routes.

6. Conclusion

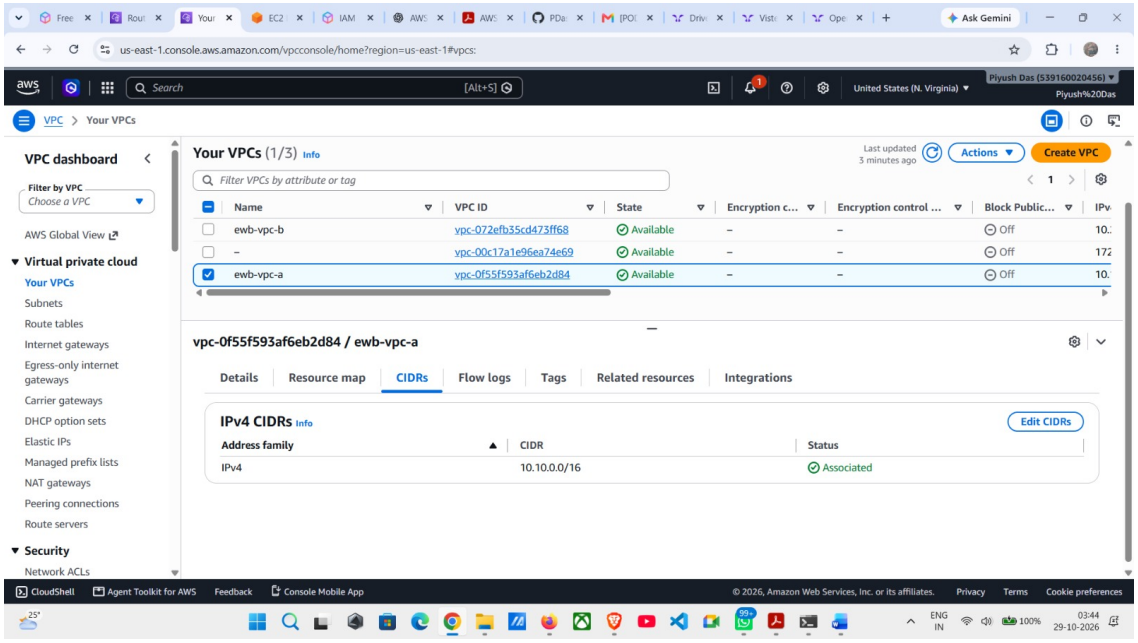
The lab demonstrates VPC Peering between two VPCs with non-overlapping IPv4 CIDR blocks, reciprocal route-table entries, and private-instance connectivity. The screenshots included below provide supporting evidence of the AWS configuration and connectivity tests.

Operational note: After preserving this report and all required evidence, review and remove lab resources that are no longer needed to avoid ongoing AWS charges. Resource cleanup is a separate console task and is not claimed as completed in this report.

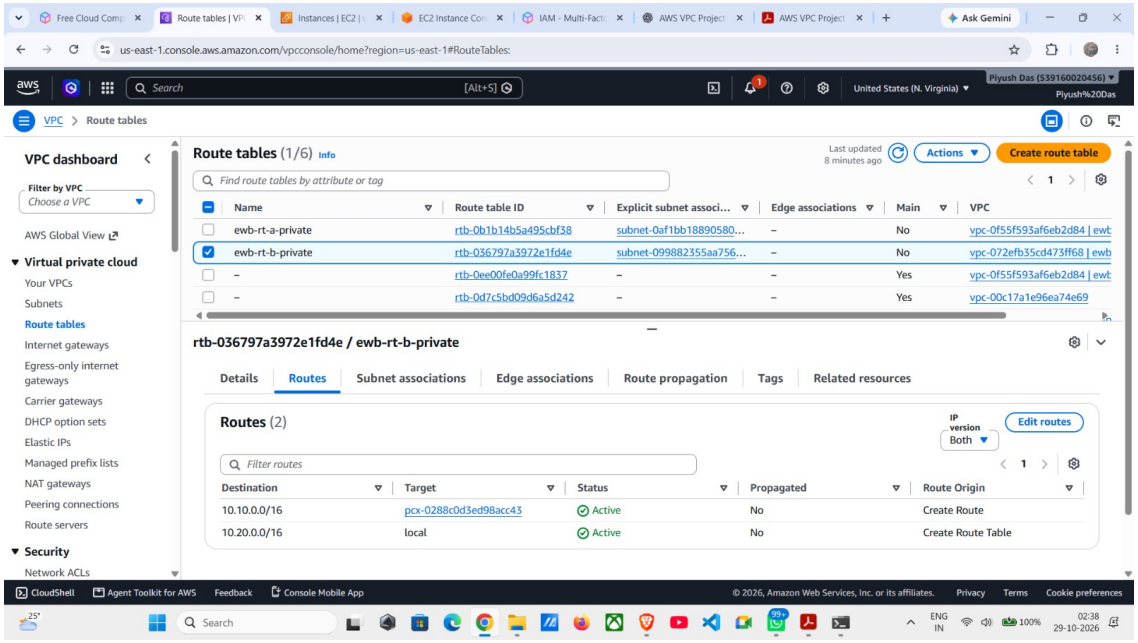
7. Screenshots and Evidence

The following screenshots were supplied during the project walkthrough.

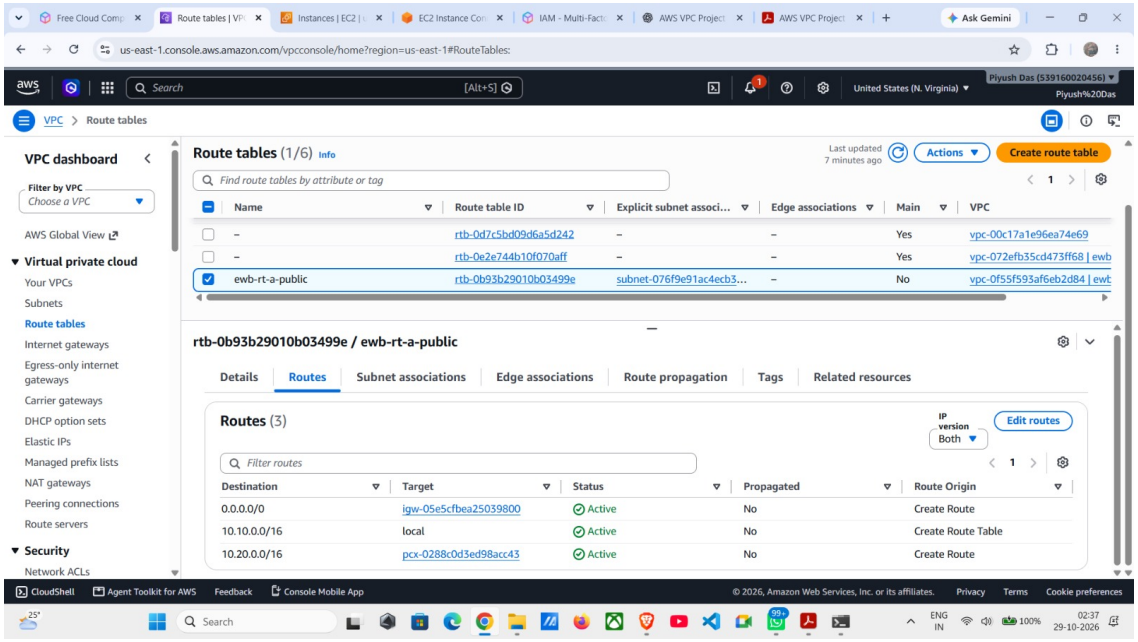
VPC A and VPC B CIDR configuration



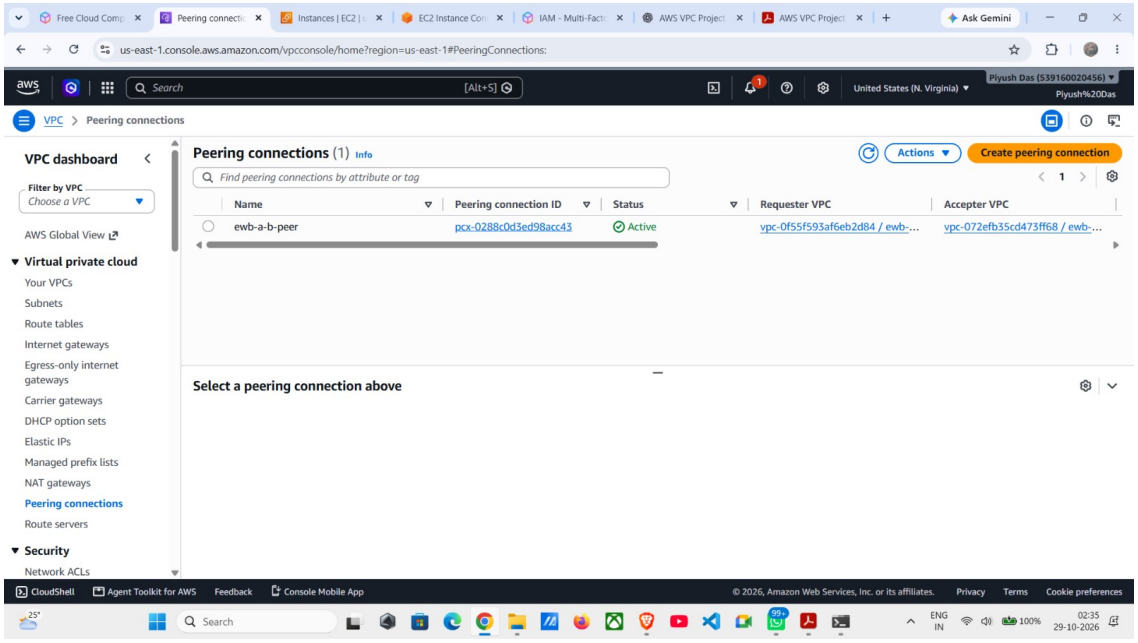
VPC A private route table: route to VPC B through peering



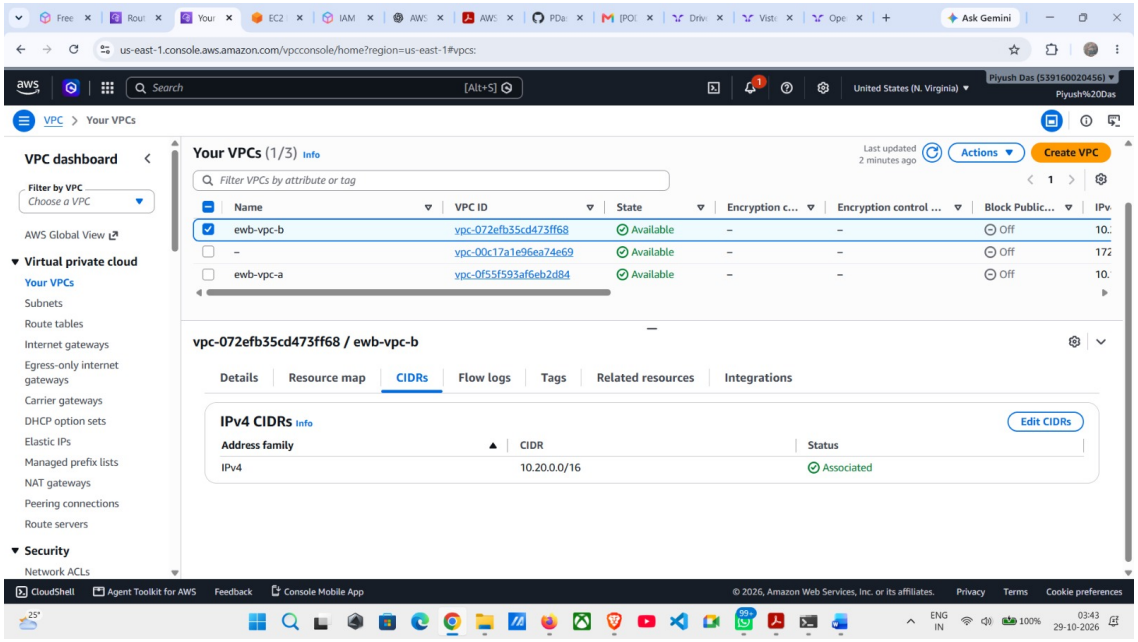
VPC B private route table: route to VPC A through peering



Active VPC peering connection



Connectivity test: App-B to App-A, 4 received and 0% packet loss



Connectivity test through the bastion host

